III.6 Degenerovaný Fermiho plyn

→ nerelativistické neinteragující fermiony (vodivostní pásy v kovech)

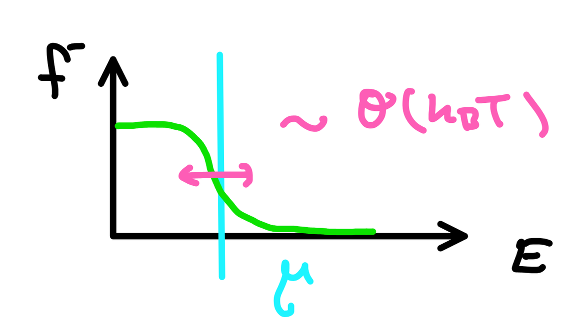
- silně kvantový režim $\rho\Lambda^3 \gtrsim 1$
- spin $1/2$ ⇒ degenerace 2
- hustota: $\rho = \frac{1}{V}\sum_k f_k^- \to \frac{1}{V}\int dE\, \gamma(E)\, f^-(E)$

$$\xrightarrow{V\to\infty} \Lambda^{-3}(T)\, \underbrace{g^-_{3/2}(z)}$$

konverguje pro $|z|\le 1$ $\quad = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n^{3/2}}$

- ale: $\mu > 0$ je fyz. možné ⇒ $z > 1$ je možné

⇒ suma nekonverguje, je nutné použít Sommerfeldův rozvoj kolem $T = 0$



→ Fermi-Diracova statistika:

$$f^-(E) = \frac{1}{1+e^{\beta(E_k-\mu)}} = \frac{1}{2} - \frac{1}{2}\tanh\frac{E_k-\mu}{2k_BT} \xrightarrow{T\to 0^+} \Theta(\mu - E)$$

$\boxed{T=0}$

→ jsou zaplněny všechny stavy do μ

$$\rho = \frac{1}{V}\int_0^{\mu} \gamma(E)\, dE = \frac{1}{V}\underbrace{\Gamma(\mu)}_{V\cdot g\cdot \frac{1}{h^3}\frac{4}{3}\pi(2mE)^{3/2}\big|_{E=\mu}}$$

→ Fermiho energie: $E_F = \mu\big|_{T=0}$

$$E_F = \frac{\hbar^2}{2m}(3\pi\rho)^{2/3} \propto \rho^{2/3}$$

- Fermiho teplota: $T_F = E_F / k_B$
- Fermiho hybnost $p_F = \hbar(3\pi\rho)^{1/3}$

→ tlak: $p = \frac{1}{V}\int_0^{\mu} \Gamma(E)\, dE = \frac{2}{5}\rho E_F \propto \rho^{5/2}$

→ hustota energie $u = \frac{3}{2}p = \frac{3}{5}\rho E_F \propto \rho^{5/2}$

$\boxed{k_BT \ll E_F}$

→ obecná situace:

$$I = \int_0^{\infty} \phi(E) f^-(E)\, dE \overset{PP}{=} \underbrace{\left[\psi f^-\right]_0^{\infty}}_{0} - \int_0^{\infty} dE\, \psi \frac{\partial}{\partial E}\left[\frac{1}{2} - \frac{1}{2}\tanh\frac{E-\mu}{2k_BT}\right]$$

$\phi(E) = \frac{d\psi}{dE}$

$\psi(0) = 0$

$$= \int_0^{\infty} \psi(E)\, \frac{1}{4}\beta \frac{1}{\cosh^2\frac{\beta(E-\mu)}{2}}\, dE$$

Taylor kolem $E = \mu$

$$= \frac{\beta}{4}\sum_{n=0}^{\infty} \psi^{(n)}(\mu) \int_0^{\infty} \frac{(E-\mu)^n}{n!} \frac{1}{\cosh^2\frac{\beta(E-\mu)}{2}}$$

$$= \sum_{n=0}^{\infty} \frac{1}{4\beta^n}\psi^{(n)}(\mu) \int_{-\beta\mu}^{\infty} \frac{x^n}{n!}\frac{1}{\cosh^2\frac{x}{2}}$$

$$\approx \sum_{n=0}^{\infty} \frac{\psi^{(n)}(\mu)}{n!}(k_BT)^n \underbrace{\int_{-\infty}^{\infty} \frac{x^n}{4\cosh^2\frac{x}{2}}}$$

$= 0$ pro liché n
$= 1$ pro $n = 1$

$$\int_0^{\infty} \phi(E) f^-(E)\, dE = \int_0^{\mu} \phi(E)\, dE + \sum_{n=1}^{\infty} \frac{J_n}{(2n)!}\phi^{(2n-1)}(\mu)(k_BT)^{2n}$$

- speciálně $J_1 = \frac{\pi^2}{3}$

<u>Korekce nejnižšího řádu</u>

$$\rho = \frac{1}{V}\int_0^{\infty} \gamma(E) f^-(E) = \underbrace{\frac{1}{V}\int_0^{\mu}\gamma(E)\, dE}_{\frac{1}{V}\Gamma(\mu) = A\cdot\mu^{3/2} = \rho\left(\frac{\mu}{E_F}\right)^{3/2}} + \frac{\pi^2}{6V}\underbrace{\gamma'(\mu)}_{\Gamma''(\mu) = \frac{3}{2}\cdot\frac{1}{2}AV\mu^{-1/2}}(k_BT)^2$$

$$= \rho\left(\frac{\mu}{E_F}\right)^{3/2} + \frac{\pi^2}{8}\left(\frac{\mu}{E_F}\right)^{-1/2}\left(\frac{k_BT}{E_F}\right)^2$$

vyjádříme

$$\frac{\mu}{E_F} = 1 - \frac{\pi^2}{12}\left(\frac{k_BT}{E_F}\right)^2 + O\left(\left(\frac{k_BT}{E_F}\right)^4\right)$$

→ tlak

$$p = \frac{1}{V}\int_0^{\infty}\Gamma(E) f^-(E)\, dE = \frac{1}{V}\int_0^{\mu}\underbrace{\Gamma(E)}_{A\cdot V\cdot E^{3/2}}\, dE + \frac{\pi^2}{6}\frac{1}{V}\underbrace{\gamma(\mu)}_{A\cdot V\cdot\frac{3}{2}\mu^{1/2}}(k_BT)^2$$

$$= \frac{2}{5}\rho E_F\left(\frac{\mu}{E_F}\right)^{5/2} + \frac{\pi^2}{4}\rho\left(\frac{\mu}{E_F}\right)^{1/2}\left(\frac{k_BT}{E_F}\right)^2 E_F$$

$$= \frac{2}{5}\rho E_F\left[\underbrace{\left(\frac{\mu}{E_F}\right)^{5/2}}_{1-\frac{5}{2}\frac{\pi^2}{12}\left(\frac{k_BT}{E_F}\right)^2} + \frac{5}{8}\pi^2\underbrace{\left(\frac{\mu}{E_F}\right)^{1/2}}_{\approx 1}\left(\frac{k_BT}{E_F}\right)^2\right]$$

$$p = \frac{2}{5}\rho E_F\left[1 + \frac{5\pi^2}{12}\left(\frac{k_BT}{E_F}\right)^2\right]$$

→ $u = \frac{3}{2}p = \frac{3}{5}\rho E_F\left[1 + \frac{5\pi^2}{12}\left(\frac{k_BT}{E_F}\right)^2\right]$

$$c_V = \frac{\partial u}{\partial T} = \frac{1}{2}k_B\rho\frac{k_BT}{E_F} \propto T$$